KISAAN SAATHI

Intelligence for Every Acre

Harnessing AI & Space Tech for Smallholder Farmers



NOVEMBER 2026

Can India’s AI and Space Intelligence Reach the People Who Feed Us?

India is investing heavily in AI and space technology; can that intelligence reach the small farms that feed us?



Small farms. High-stakes decisions.

1.08 ha

Average operational holding size
Agriculture Census 2015–16

Fragmented information turns every crop, soil, weather and market choice into a complex decision.



India already monitors agriculture from space — FASAL

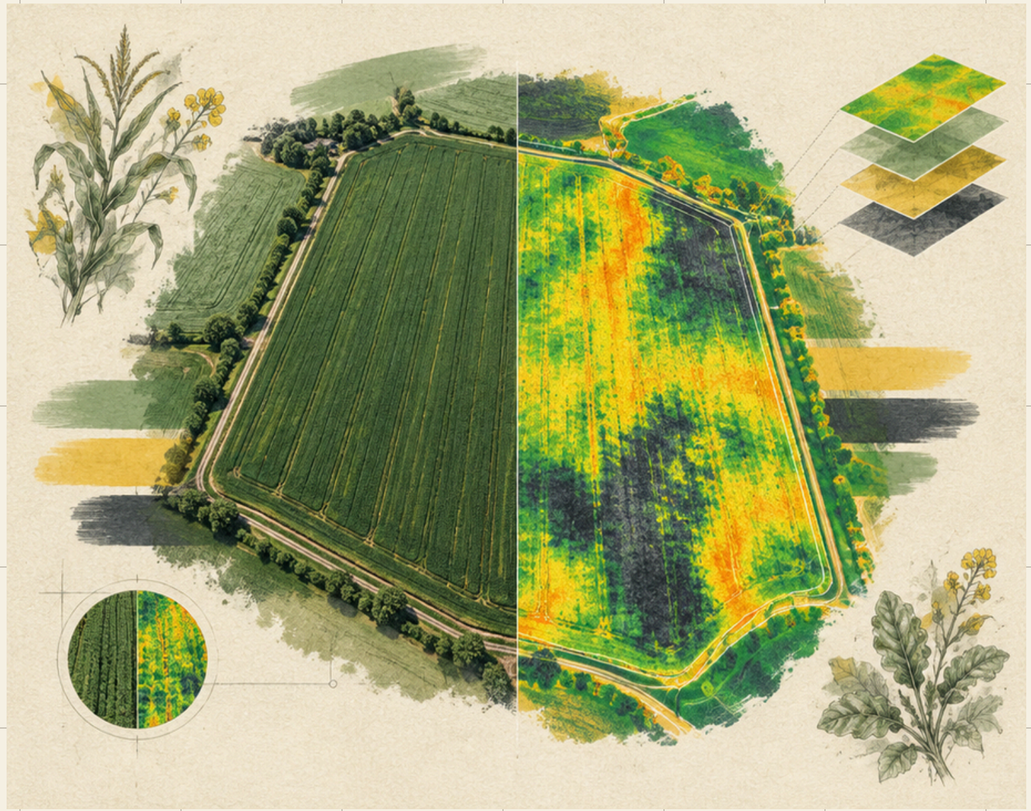
FASAL integrates satellite data, agrometeorology and field inputs.

20 states / 557 districts

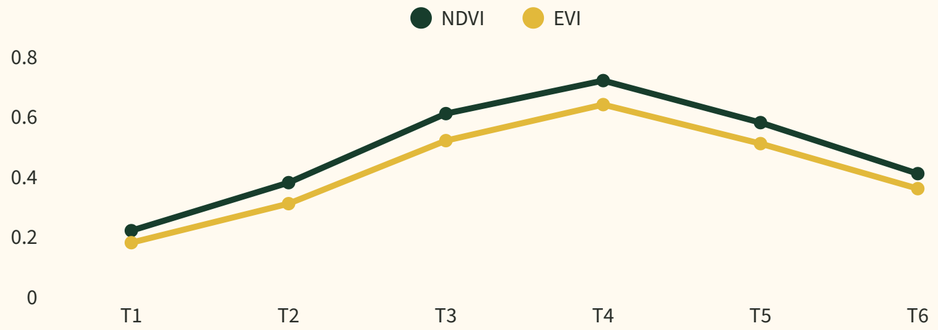
Scope reported by PIB, December 2025



A vegetation signal is not a diagnosis.



Sentinel-2 • 10–20 m bands



Useful: time series + field boundaries
Limit: mixed pixels at small-field edges
Interpretation requires ground context

From a field boundary to an evidence-backed next step

1 Select or draw field

Satellite • weather • soil • crop • market

2 Assemble context

Evidence + assumptions

3 Explain reasoning

Confidence level

4 Next steps / report

What to verify on ground

The recommendation should show its evidence, assumptions, confidence — and what still needs eyes on the field.

Proposed PDF intelligence workflow

PROPOSED • NOT IMPLEMENTED



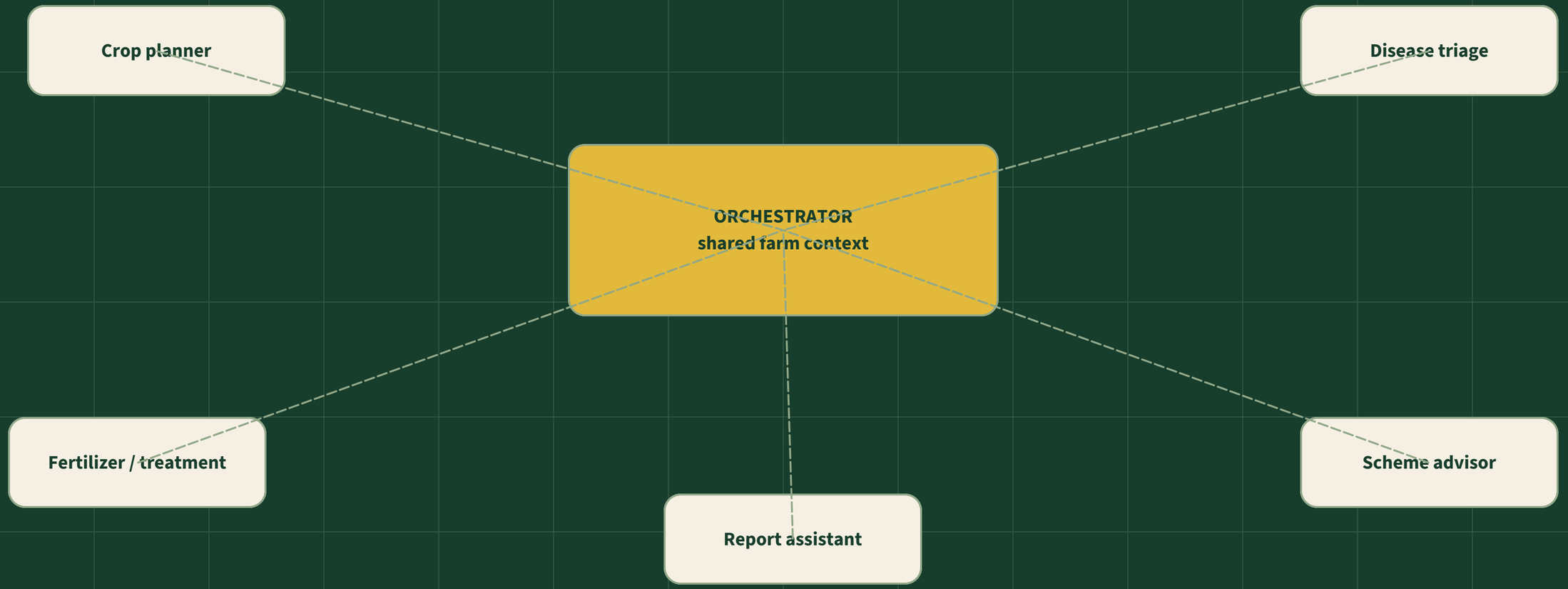
Cite page + section

State uncertainty

No evidence? No fabrication.

A proposed team of specialist agents

PROPOSED ARCHITECTURE



Memory and permissions are different jobs

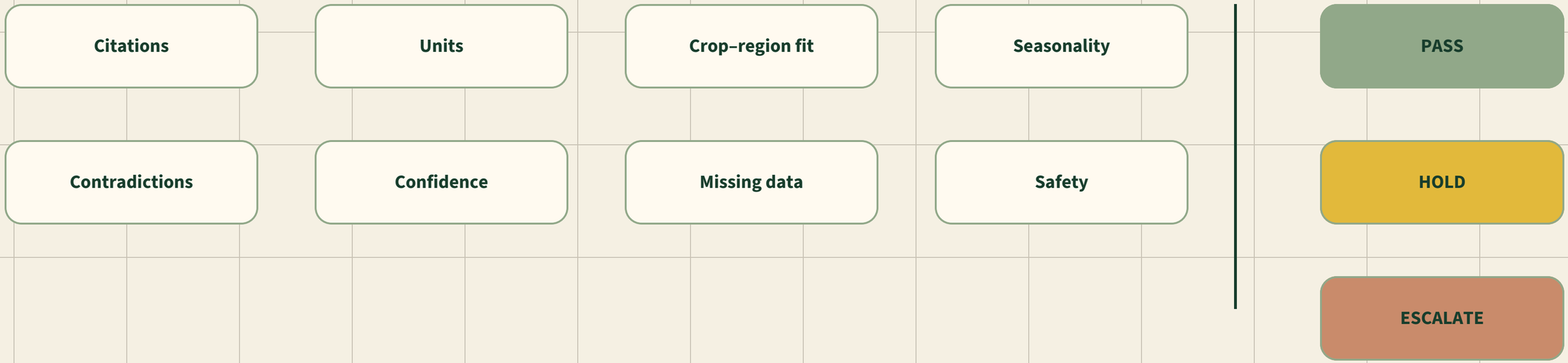
AGENTS — Specialist reasoning

MEKO — Shared memory / context per farmer + farm

MCP — Permissioned tools + resources — not a memory store

DATABASE / CHECKPOINTER — Persistent state with explicit scoping

A quality gate — not a guarantee



Flags issues for review. It does not certify agronomic correctness.

Build the spine first. Connect evidence carefully.

IMPLEMENTED SCAFFOLD

- PWA shell
- Health endpoint
- Schemas
- Agent / RAG modules
- Tests + CI workflow

PROPOSED / UNIMPLEMENTED

- Provider connections
- Persistence + MCP tools
- PDF ingestion / retrieval
- Evaluation + field validation
- Usability testing

Implementation status, roadmap, testing & trust

01

SCAFFOLD

PWA • endpoints • schemas • modules • tests • CI/CD

02

CONNECT

Providers • persistence • MCP • PDF retrieval

03

EVALUATE

Evidence quality • safety • usability • field validation

04

DEPLOY

Intended rollout after validation — not live today

Trust is measured through transparency, traceability, review quality and iterative learning — without invented performance numbers.

SOURCES

Agriculture Census 2015–16

PIB, December 2025

Kisaan Saathi Capstone PDF — cited pages / sections

“The next revolution in farming cannot just be about growing more. It has to be about knowing more.”

Enabling farmers with technology to transform decisions and lives.

